

Learning a Common Index from Outcomes Measured on Non-Comparable Scales

Project proposal · Nicolas Abbate · October 5, 2026

1 Motivation

Aerial imagery now covers most of the United States at building-level resolution and annual frequency, while the American Community Survey reports five-year averages for census tracts with margins of error that are wide for small areas. A model that reads wealth from imagery could close that gap. It would help target programs at a geography the survey cannot support, sharpen small-area estimates by combining the two signals, and measure change between survey rounds. It would also make impact evaluation possible where no survey covers the affected places or years: after a rezoning or a transit extension, the units of interest are blocks and the window is shorter than a survey cycle.

Remote wealth estimation has a fundamental problem that remains unresolved. Detection models scale because a building is a building in Dallas and in Lagos, so every labeled image is one more example of a single relationship. Wealth labels lack that property. Surveys report wealth in dollars, and the same visual bundle carries different dollar values in two cities or in two years, because each local wealth distribution is shifted, stretched and reshaped by forces unrelated to the built environment. Pooling dollar labels across places and periods therefore mixes incompatible relationships, and a new city can add an inconsistent example rather than another instance of the same one.¹

One reading of this evidence is that the map from features to wealth is inherently local, so every city and year needs its own model. We read it differently: the locality is in the dollar labels, not in the ordering. Wealth buys amenities that look much the same everywhere—space, light, quiet, greenery, quality of construction, height, centrality. Because of this, we believe that there exists a global relation between the features and wealth; what is not global, and changes from city to city and year to year, is the number of dollars attached to each feature bundle.

The problem is not specific to imagery or to wealth. It arises wherever outcomes are comparable within groups but not across them: a fixed income affords different housing across cities, exams vary in difficulty, and credit ratings depend on the rating system. The same structure appears in preference and reward data with annotator-specific scales, test scores across forms and graders, credit risk across vintages, severity graded from medical images across sites and scanners, and hedonic quality across markets whose price levels differ. In each case covariates determine a shared index that every group reports through its own unknown monotone rescaling.

This project studies how to learn the shared index, one function of the covariates that applies in every group, without estimating the group-specific rescalings. Within a group, the labels order units as the index does. Across groups, observations known from outside the labels to be nearly the same, such as one location photographed in two years with no structural change, must receive nearly the same index, a restriction that uses no label. We combine them in one training criterion: a pairwise ranking loss within groups plus a scale-free penalty on how much the score moves over such pairs. We then prove consistency of the estimator under three assumptions on the comparison errors, and run a Monte Carlo study of when each version works, when it fails, and whether the unlabeled pairs improve prediction.

¹As an illustration, Khachiyan et al. (2022) report $R^2 \approx 0.90$ for income levels pooled over 2000 and 2010 and $R^2 \approx 0.40$ for the 2000–2010 change. While they do not work with wealth, the same label distortion problem arises in their setting.

2 Framework

Units are indexed by i , groups by c (cities) and dates by t ; the pair (c, t) is a *cell*. We observe a label $W_{ict} \in \mathbb{R}$ and covariates X_{ict} , an image tile in the application. The model is

$$W_{ict} = \xi_{ct}(X_{ict}) + e_{ict}, \quad \xi_{ct} = \psi_{ct} \circ f. \quad (1)$$

The *global function* f maps covariates into a latent index and is the same in every cell. The *cell-specific function* ψ_{ct} converts the index into the units in which the cell reports it (dollars in city c in year t) and carries that cell's level, spread and shape. A model fitted cell by cell learns ξ_{ct} , which says nothing about a cell without labels. The goal is to estimate f without estimating ψ_{ct} .

Let $\mathcal{S}$ be a set of *stable pairs* (x, x') : covariate values observed in two different cells and selected, without reference to the labels, as nearly the same unit.

Assumption A1 (Monotonicity). ψ_{ct} is strictly increasing for every (c, t) .

Assumption A2 (Invariance). f does not depend on c or t .

Assumption A3 (Regularity). There is a factorization $f = g \circ \phi$, with the coordinates of $H = \phi(X) \in \mathbb{R}^K$ standardized, such that $|g(h) - g(h')| \leq L \|h - h'\|$ on the support of H , and every stable pair $(x, x') \in \mathcal{S}$ satisfies $\|\phi(x') - \phi(x)\| \leq \delta$. The index is not constant within cells.

A2 is the substantive assumption: all variation across cells is assigned to ψ_{ct} . In A3, H is the embedding before the final scalar layer of a network, and L and δ matter only through their product.

Proposition 1 (Pooled levels estimate a mixture). Suppose (1) holds with $\mathbb{E}[e_{ict} | X_{ict}, c, t] = 0$ and $\mathbb{E}|W| < \infty$, and W is regressed on X pooling all cells. Then

$$\mathbb{E}[W | X = x] = \sum_{c,t} \Pr(c, t | X = x) \psi_{ct}(f(x)).$$

The weights reflect which cells are in the sample, and they can reverse the ranking. If public housing towers appear only in New York and single-family houses mostly in Dallas, a large enough dollar gap between the cities makes the pooled fit rank the towers above the houses, although they are poorer in each city: the model has learned that public housing signals New York.

Proposition 2 (What the data reveal about f). Let A1–A3 hold.

- (i) Ordering within a cell. For every cell (c, t) and any x, x' : $\xi_{ct}(x') > \xi_{ct}(x) \iff f(x') > f(x)$.
- (ii) Stability across cells. For every stable pair $(x, x') \in \mathcal{S}$, with x observed in cell (c, t) and x' in cell $(c', t') \neq (c, t)$: $|f(x') - f(x)| \leq L\delta$.

Proof. (i) $\xi_{ct} = \psi_{ct} \circ f$ and ψ_{ct} is strictly increasing by A1. (ii) By A2 the same $f = g \circ \phi$ applies in both cells, so $|f(x') - f(x)| \leq L \|\phi(x') - \phi(x)\| \leq L\delta$ by A3. $\square$

Part (ii) holds for any two cells, so a stable pair may join two dates of one group ($c = c', t \neq t'$) or two different groups ($c \neq c'$). In satellite and aerial imagery only the first kind arises on its own: one location, two years, no recorded structural change. Nothing physical ties a tile in one city to a tile in another, so cross-city pairs require an already pretrained model, in whose representation near neighbours can be defined before training.

Proposition 2 establishes what is observable about f : orderings within cells and near-equalities across cells, neither involving ψ_{ct} . But the labels are W , not $\xi_{ct}(X)$: whether a noisy comparison still reveals the ordering of f depends on the errors, and the assumption made about them determines the loss. For two units $i \neq j$ of one cell write $y_{ij} = \mathbf{1}\{W_j > W_i\}$, $p_{ij} = \Pr(y_{ij} = 1 | X_i, X_j, c, t)$ and $\Delta_{ij}s = s(X_j) - s(X_i)$ for a score function s .

Assumption A4 (Comparison errors, one of three versions).

- (a) Median zero. *Given (X_i, X_j, c, t) , $e_i - e_j$ has median zero and a distribution function strictly increasing at zero. By Proposition 2(i), $p_{ij} > \frac{1}{2} \iff \Delta_{ij}f > 0$.*
- (b) Heterogeneous logistic. *$p_{ij} = \Lambda(\rho_{ct} \Delta_{ij}f)$ with $\rho_{ct} > 0$ and $\Lambda(z) = 1/(1 + e^{-z})$.*
- (c) Homogeneous logistic. *Version (b) with $\rho_{ct} = \rho$ in every cell.*

Version (a) restricts only the sign of $p_{ij} - \frac{1}{2}$ and lets the noise depend on the features; (b) and (c) also fix its shape, with and without a reliability that differs by cell.² All three lead to one criterion,

$$Q_k(s, \eta) = Q_k^{\text{cross}}(s, \eta) + \lambda_t \frac{S_t(s)}{V(s)} + \lambda_c \frac{S_c(s)}{V(s)}, \quad k \in \{a, b, c\}, \quad (2)$$

where $S_t(s)$ and $S_c(s)$ are the mean squared score differences $\mathbb{E}[(s(x') - s(x))^2]$ over temporal and cross-group stable pairs, and $V(s) = \frac{1}{2} \mathbb{E}[(\Delta_{ij}s)^2]$ is the within-cell dispersion of the score. Dividing by V makes the penalties scale-free, so shrinking the scores cannot satisfy them; a weight is zero when that kind of pair is unavailable. The criterion uses the labels only through y_{ij} , so it is unchanged by any strictly increasing relabeling within a cell: it never sees ψ_{ct} .

Only the first term depends on the error assumption. Under median zero an ordinal loss makes sense: the probability of misordering a pair, $Q_a^{\text{cross}}(s) = \Pr((2y_{ij} - 1)\Delta_{ij}s < 0) + \frac{1}{2} \Pr(\Delta_{ij}s = 0)$, trained through a smooth surrogate. Under heterogeneous logistic errors it is a RankNet loss parametrized at the city-year level, $Q_b^{\text{cross}}(s, \eta) = \mathbb{E}[\ell(\rho_{ct} \Delta_{ij}s, y_{ij})]$ with $\ell(z, y) = \log(1 + e^z) - yz$, $\eta = \{\rho_{ct}\}$ and the scale of s fixed by $V(s) = 1$. Under homogeneous logistic errors it is the RankNet loss of Burges et al. (2005), $Q_c^{\text{cross}}(s) = \mathbb{E}[\ell(\Delta_{ij}s, y_{ij})]$, with the scale absorbed into s .

3 Related work

Rank estimation removes an unknown monotone transformation by comparing pairs (Han, 1987), and Abrevaya (2000) lets it differ across groups; both keep a finite-dimensional index. Wang et al. (2024) place a deep network inside a rank objective, with one transformation and independent observations. In learning to rank, Burges et al. (2005) introduce the pairwise logistic loss, and Duchi et al. (2010) show that commonly used surrogates can be inconsistent for the ranking risk, which is why we compare an ordinal loss with the logistic ones. The stability terms relate to manifold regularization (Belkin et al., 2006), slow feature analysis (Wiskott and Sejnowski, 2002) and self-supervised invariance objectives (Bardes et al., 2022; Jean et al., 2019); neither a pair penalty nor its normalization is new in isolation. Invariant risk minimization (Arjovsky et al., 2019) also seeks one predictor across environments but takes labels to be on a common scale. Wealth prediction from imagery fits one model per cell (Jean et al., 2016) or pools dollars (Khachiyan et al., 2022), as in Proposition 1.

4 Intended results and evaluation

Theory. The estimator is $\hat{f} = s_{\hat{\theta}}$, where $\hat{\theta}$ minimizes the sample analogue of (2) over a neural network class. Groups are independent clusters with arbitrary dependence inside each, and asymptotics are in the number of groups. We intend to prove that $\hat{f}$ is consistent for f under each version of A4: first what the minimizers of the population criterion recover, then uniform convergence of the sample criterion. What consistency means differs by version. Under (c) the index is recovered up to scale and a constant per cell (a preliminary proof exists). Under (b) the same holds jointly with the scales ρ_{ct} , whose number grows with the sample: the main technical difficulty. Under (a) only the ordering is recovered, so excess misranking risk vanishes up to the smoothing error of the surrogate.

²Versions (b) and (c) place the noise on the scale of the index. They hold when $W = \Psi_{ct}(f(X) + \varepsilon)$ with logistic differences of ε , a model whose regression form is (1) if ε is independent of X within a cell; with logistic differences of the additive e they hold when ψ_{ct} is affine.

Monte Carlo. The design follows Morris et al. (2019) and Kiviet (2012): aims, data-generating processes, estimands, methods and performance measures are fixed before any run, every method is applied to the same simulated datasets, and one factor moves at a time from a common baseline. In that baseline the covariates have three blocks: signal that enters f , acquisition noise that is redrawn at every date, and a group cue that reveals the group; neither of the last two enters f . Labels are generated at the unit level and then transformed by ψ_{ct} , never as independent pair outcomes. Every f is rescaled so that $V(f) = 1$ and the best attainable misranking rate is equal across designs, which holds the difficulty of the task fixed.

Question	What varies	What is held fixed
Q1. Efficiency against misspecification of ψ_{ct}. When a parametric correction $\psi_{ct}^{-1}(W)$ is feasible, what does it gain if correct and lose if wrong?	Shape of ψ_{ct} along one dial (common, cell multiplier, location and scale, shape); cross-cell differences in the distribution of $f(X)$; number of labeled groups. Competitors: pooled levels, logs with cell effects, within-cell z -scores and percentile ranks, and an oracle fitted to the latent index.	f ; links; errors at (c), where our loss and a correct level regression are both well specified.
Q2. Misspecified errors. How does each of the three losses behave when its version of A4 fails?	One departure at a time, each with one dial: tail weight of $e_i - e_j$; dispersion of ρ_{ct} across cells; dependence of the noise scale on X ; skewness with a scale that depends on X , which violates (a).	ψ_{ct} ; f ; sample size; links.
Q3. Value of stable pairs. Within each loss, do the links improve on a well-trained fit without them?	Link arm: none, temporal only, cross-group only, both. Strength of the acquisition noise and of the group cue.	Labels; architecture; tuning budget; errors; ψ_{ct} .
Q4. Approximate and false links. How fast do gains erode, and does tuning the penalty weight prevent harm?	A drift in f on every link; a share of entirely false links, random or one-sided; penalty weight fixed or chosen on held-out ranking loss.	The design of Q3 with both kinds of link.

Table 1: Monte Carlo questions, each run for the three losses and for a linear and a nonlinear f .

Q1 and Q3 are read from held-out error against the number of labeled groups: a correct competitor appears as a horizontal gap, the extra labels our method needs, and a misspecified one as a floor that more data does not remove. Q2 and Q4 are read from error against the dial.

Performance measures. All are computed in held-out groups and reported with Monte Carlo standard errors: (i) the share of within-cell pairs ordered differently from the true f , comparable across the three losses; (ii) the same share for pairs across dates and across groups, which a simulation allows because f is known; and (iii) error in within-cell index differences with $V = 1$, for the logistic losses only, since the ordinal loss identifies no spacing.

5 Execution plan and expected contribution

The work has five steps. (1) Derive the population result for the three versions of A4. (2) Build the simulator and the three losses from scratch, including the smooth ordinal surrogate and the minibatch estimate of the ratio penalties, with a linear head on fixed features and with a small network. (3) Run Q3 first: if links do not help against a well-trained fit without them, that is the finding we report. (4) Run Q1, Q2 and Q4. (5) Prove sample consistency for the versions that survive and formalize the regularities the simulations show. Time permitting, we illustrate the method on U.S. aerial imagery and tract incomes.

We expect three contributions: a training criterion that learns one predictor from labels comparable only within groups, invariant to the label scale by construction; consistency results

that state what each error assumption identifies, from an ordering to an index with cell-specific reliabilities; and evidence on when unlabeled stable pairs improve prediction, when they harm it, and what is lost relative to a correct parametric correction.

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